Flag relations, reconstruction and frame orbits

Human Compression project

11 October 2026 — author draft

Scope and attribution

This paper supplies a reusable interface for triangle constructions on finite linear spaces. The construction is due to Delaby, Leemans and Tranchida [DLT, Construction 1]. The common-neighbor reconstruction, correlation extension and deleted-pencil description below reuse the project's earlier `incidence_compression.md`, §4. We give complete proofs so that the present paper is an actual dependency of an application, rather than a collection of references to an old combined manuscript. The point-fiber synchronization in Theorem 2 also covers two-point lines. The determinant criterion in Theorem 5 makes explicit the semilinear alternative omitted from DLT's affine-plane classification. Neither construction nor classical linear algebra is claimed as a new subject. The accompanying application identifies precisely which source statements these arguments recover or correct.

All sets in the geometric part are finite. A linear space S has a point set P, a line set L, at least two points on each line, at least two lines through each point, and a unique line through any two distinct points. A flag is an incident pair (p,L). A uniform space has k points on each line and r lines through each point; consequently v−1=r(k−1), where v=|P|. We use "automorphism" for a type-preserving incidence automorphism and "correlation" when interchange of points and lines is allowed. A duality need not have order two; a polarity does.

1. A directed relation on flags

Write F for the flag set. Define the relation D by

$$(p, L)D(q, M) \quad \Longleftrightarrow \quad p \in M \text{ and } q \notin L. \tag{1}$$

Because q belongs to M, the conditions imply p≠q and M=pq≠L. If A and B are the zero-one matrices for equality of point and equality of line on F, then

$$D = (A - I)(B - I). \tag{2}$$

Indeed the only possible intermediate flag is (p,M). This observation also proves that the integer product has entries zero or one.

The triangle incidence system Δ(S) has elements (f,i), where f∈F and i∈Z/3Z. Distinct types i,i+1 are incident when fDg, respectively; the reverse incidence is determined by symmetry. Its adjacency matrix, in cyclic type order, is

$$\begin{pmatrix} 0 & D & D^{\mathsf{T}} \\ D^{\mathsf{T}} & 0 & D \\ D & D^{\mathsf{T}} & 0 \end{pmatrix}. \tag{3}$$

The cyclic shift τ(f,i)=(f,i+1) is therefore a triality. If p0,p1,p2 are an ordered noncollinear triple, its chamber has flag of type i equal to (pi,p(i−1)pi), with subscripts read cyclically. Conversely, (1) forces these three joining lines in any chamber and excludes collinearity. A panel between (p,L,0) and (q,pq,1) has exactly one completion for each a∈L{p}: the third flag is (a,qa,2). Thus every flag extends to a chamber, and every panel has k−1 completions in the uniform case. In particular Δ(S) is firm precisely when k≥3 and is thin precisely when k=3.

For a general rank-two input, the existence of a chamber of the literal triangle system already forces a six-cycle in the input incidence graph: its three point labels and three line labels are distinct, consecutive incidences come from the chamber, and the required nonincidences prohibit repeated labels. Consequently, if that nonempty triangle system is a geometry, the input has gonality at most three. This recovers DLT Proposition 3.3; it does not assert that every input of such gonality gives a geometry.

The outgoing degree of the flag (p,L) is v−|L|, since each q outside L gives the unique successor (q,pq). Hence flag transitivity of Δ(S) forces every line of S to have the same size k: its automorphism group is transitive on each vertex type. The identity v−1=r_p(k−1), obtained by partitioning P{p} among the lines through p, then forces a constant r_p=r. This proves uniformity without assuming that the automorphisms of Δ(S) have already been reconstructed.

2. Reconstruction from common neighbors

Lemma 1 (point and line signatures). For distinct flags f,g in a uniform linear space, the following alternatives hold:

Relation between f and g	Common successors	Common predecessors
same point	(r−2)(k−1)	not needed
different points	0 or k−2	not needed
same line	not needed	(k−2)(r−1)
different lines	not needed	0 or r−2

For two flags (p,L),(p,M) at the same point, a common successor uses a line through p different from L and M, and then one of its k−1 other points. For flags at different points p,q, its line must be pq. If neither forbidden line is pq, its point may be any of the k−2 points on pq other than p and q; otherwise there are no choices. Reversing the reasoning proves the last two rows. Distinct lines either meet in one point or do not meet, so the reverse count has no further possibilities. This proves the lemma.

Theorem 2 (type-preserving reconstruction). If r≥3 and k≥2, restriction to the underlying flags identifies Aut(Δ(S)) with Aut(S).

The value (r−2)(k−1) is strictly larger than k−2. Thus the maximum off-diagonal common-successor relation on each type identifies precisely which flags have the same point. Every type-preserving automorphism φ of Δ therefore induces a permutation fi of P on type i. Write

$$\phi(p, L, i) = (f_i(p), M_i(p, L), i).$$

The point labels occurring among the successors of (p,L,i) are exactly P\L. Applying φ gives

$$P \setminus M_i(p, L) = f_{i+1}(P \setminus L), \qquad M_i(p, L) = f_{i+1}(L). \tag{4}$$

Since the image is a flag, fi(p) belongs to every f(i+1)(L) with L through p. At least two distinct lines pass through p and their intersection is {p}. Intersecting these images shows fi(p)=f(i+1)(p). All three

point permutations are therefore a single permutation f, and (4) shows that f maps every line to a line and that φ is its diagonal action on flags. The inverse satisfies the same condition, so f is an automorphism of S. Conversely every automorphism of S preserves (1), proving the claimed equality. The same argument, with the target point set written P′, applies to any isomorphism Δ(S)→Δ(S′) between uniform spaces with the same parameters k,r: equation(4) becomes P′\M_i=f_(i+1)(P\L), and the same intersections synchronize the three point bijections. Thus every such triangle isomorphism is induced by one unique isomorphism S→S′. Notice that the proof never needs k−2>0; two-point lines are included.

If k,r≥3, both exceptional common-neighbor values in Lemma 1 are strict maxima. A correlation reversing the cyclic type order exchanges the successor and predecessor signatures and therefore exchanges the two flag partitions. On each type it therefore induces a bijection from points to lines and from lines to points. A point fiber and a line fiber meet in a flag exactly when their labels are incident, so these two bijections form a duality δi of S. In particular the action on type 0 is induced by one duality δ0. The globally defined lift of δ0 reverses the cyclic type order, since it transposes D. Composing the given correlation with its inverse lift gives a type-preserving automorphism which is the identity on type 0. Theorem 2 forces that automorphism to be the identity on every type. Thus all three actions were induced by δ0. No extra pointwise kernel has been suppressed in this reconstruction.

Theorem 3 (the complete correlation extension). For k,r≥3 there is a natural isomorphism

$$\mathrm{Cor}(\Delta(S)) \simeq C_3 \rtimes_\epsilon \mathrm{Cor}(S), \tag{5}$$

where ε records whether a correlation preserves or interchanges the two types of S, and ε=−1 acts on C3 by inversion. If S has no duality, (5) is C3×Aut(S). If S has a duality, the different exact sequence with kernel Aut(S) and quotient S3 splits if and only if S has a polarity.

For completeness, a correlation δ of S acts on a flag by δ(p,L)=(δp,δL) when it preserves types and by (δL,δp) when it interchanges them. Define

$$\widetilde{\delta}(f,i) = (\delta f, \epsilon(\delta) i).$$

This is a homomorphism into Cor(Δ), fixes type 0, and conjugates τ to τ raised to ε(δ). The reconstruction just proved shows that τ and all these lifts generate the full correlation group. The two displayed factors have trivial intersection, proving (5). A polarity supplies an involutory lift of a type transposition and, together with τ, a complement S3. Conversely any complement has such an involutory lift, whose induced duality of S has order two. This proves the splitting criterion. A polarity is not required for (5) itself; that qualification belongs only to the latter exact sequence.

The chamber correspondence and Theorem 2 give the useful equivalence

$$\Delta(S) \text{ is flag-transitive} \quad \Longleftrightarrow \quad \mathrm{Aut}(S) \text{ is transitive on ordered noncollinear triples.} \tag{6}$$

The same assertion holds for any specified subgroup of Aut(S), using its induced diagonal action.

3. Connectivity and the deleted-pencil residue

For k,r≥3, Δ(S) is connected. In a fixed type, flags with the same point have a common successor and flags with the same line have a common predecessor, by Lemma 1. These moves connect all flags because the incidence graph of a linear space is connected: two points have their joining line. Thus each type lies in one component, and an edge between successive types joins the three components.

Theorem 4 (residue interface). The residue at (p,L,0) is the bipartite incidence structure with parts

$$\{Q : Q \notin L\}, \qquad \{(a, M) : a \in L \setminus \{p\},\ M \neq L,\ a \in M\}, \tag{7}$$

and incidence Q∈M. For the first part, the corresponding original vertex is (Q,pQ,1); the second part already gives the original vertex of type 2. The residue in the Van Nypelseer construction used in L092 has precisely the same description with a allowed to range over all of L. Formula (7) is obtained by deleting one pencil, not by assuming that this deletion preserves connectivity.

To verify (7), incidence with (p,L,0) forces a type-1 vertex to be (Q,pQ,1) for Q outside L, and a type-2 vertex to be (a,M,2) with a∈L{p} and M≠L. Their mutual incidence requires Q∈M and a∉pQ. The latter condition is automatic, since a≠p and pQ meets L only in p. This proves both directions and the claimed literal identification.

In projective or affine dimension at least three, each edge in (7) stays inside the plane generated by L and Q. Different such planes have disjoint outside-point sets and disjoint allowed-line sets. Thus the residue is a disjoint union of its plane residues. Whenever those plane residues are connected, this is its exact component decomposition.

4. Determinants classify affine-plane frame orbits

We write vectors as rows. Let K be a finite field and H a subgroup between SL(d,K) and ΓL(d,K), with d≥2. A semilinear element (A,σ) acts as v↦v^σ A. For the affine group V⋊H, translating the first point of an ordered noncollinear triple to zero identifies its orbit with the H-orbit of an ordered independent pair of vectors.

Theorem 5 (frame-orbit criterion). If d≥3, H is transitive on ordered independent pairs. If d=2, its orbits on ordered bases are in bijection with the orbits on K* of its induced determinant-semilinear transformations

$$t \longmapsto \det(A)\,\sigma(t). \tag{8}$$

In dimension two, the affine group is transitive on ordered noncollinear triples exactly when the action (8) is transitive. For a linear H≤GL(2,K) this is equivalent to H=GL(2,K). For a semilinear H it need not imply containment of GL(2,K).

For d≥3, extend either ordered independent pair to a basis. A linear map sending one complete basis to the other may have determinant c≠1. Multiply one of the target basis vectors outside the given pair by c^{-1}. The resulting map has determinant one and still sends the required pair correctly.

For d=2, arrange an ordered basis as the two rows of an invertible matrix B. If B and C have equal determinant, there is exactly one matrix M=B^{-1}C with BM=C, and det M=1. Thus SL(2,K) acts regularly on every determinant fiber. A semilinear transformation sends B to B^σ A and its determinant to (8). These facts prove both implications of the orbit correspondence: an H-orbit has one orbit of determinant values; conversely first move a determinant to the desired value and then use the contained SL group within that fiber. The transformations (8) form a group because they are induced by an actual action on matrices. For a linear H, the determinant image is a subgroup of K* acting by multiplication. Transitivity forces the full image K*; together with its contained kernel SL(2,K), this gives H=GL(2,K).

The statements about equal-determinant fibers, semilinear determinant transport, and orbit saturation are formalized in `phase265_formal/DeterminantFibers.lean`. The geometric classification, finite-field subgroup identifications and connectivity proofs are written mathematics, not consequences already formalized in that file.

5. A classical projective-line normalizer interface

We record an elementary proof of the normalizer used by the circle branch of the application, so that it is not hidden in a higher-dimensional collineation theorem. This is a classical result, not a novelty claim.

Theorem 6. For the natural action on P¹(K), where K is any finite field,

$$N_{\mathrm{Sym}(\mathbf{P}^1(K))}(\mathrm{PSL}(2,K)) = \mathrm{P\Gamma L}(2,K). \tag{9}$$

Let f normalize PSL. Since PGL is sharply three-transitive and itself normalizes PSL, composing f with a projectivity lets us suppose that f fixes ∞,0,1. The stabilizer of ∞ in PSL consists of x↦ax+b with a a nonzero square (all nonzero a in even characteristic). Its translation subgroup U={x↦x+b} is its unique Sylow p-subgroup, since U is normal and the scalar factor has order prime to p. Thus f normalizes U. Because f fixes0, conjugation sends the translation by b to the translation by f(b), giving

$$f(x+b) = f(x) + f(b). \tag{10}$$

The stabilizer of ∞ and0 in PSL consists of the square multipliers. Since f also fixes1, conjugating the multiplier by a square s gives the multiplier by f(s), and hence f(sx)=f(s)f(x) for every nonzero square s; the assertion also holds for s=0. In even characteristic every field element is a square, so f is multiplicative. In odd characteristic every t∈K is a sum of two squares: the set of squares including0 and its translate t minus that set both have (q+1)/2 elements, so they intersect. Write t=s+u with s,u squares. Additivity gives

$$f(tx) = f(sx) + f(ux) = (f(s) + f(u))f(x) = f(t)f(x).$$

Thus f is a field automorphism in all cases. Undoing the initial projectivity proves containment in PΓL. Conversely projectivities and field automorphisms normalize PSL by their matrix action, proving equality. The proof includes q=2,3: the same translation and square arguments apply, without assuming simplicity of PSL. ■

References and dependency boundary

[DLT] R. Delaby, D. Leemans and P. Tranchida, *Geometries with trialities arising from linear spaces*, Algebraic Combinatorics 9 (2026), 379–402, DOI10.5802/alco.482. Construction (1) and the geometric targets are theirs.

[IC] *Incidence compression*, repository manuscript, §4. Its flag-matrix, correlation and deleted-pencil arguments are reused above. The application must charge this complete rewritten dependency; citing it does not make its proof cost disappear.

The group classification of locally two-transitive linear spaces and the Hermitian norm-shell theorem are not proved in this umbrella. They enter only in the application and are listed there as retained inputs. A source correction, an independently verified new question answer, and a recovered principal contract have different meanings and receive separate accounting.